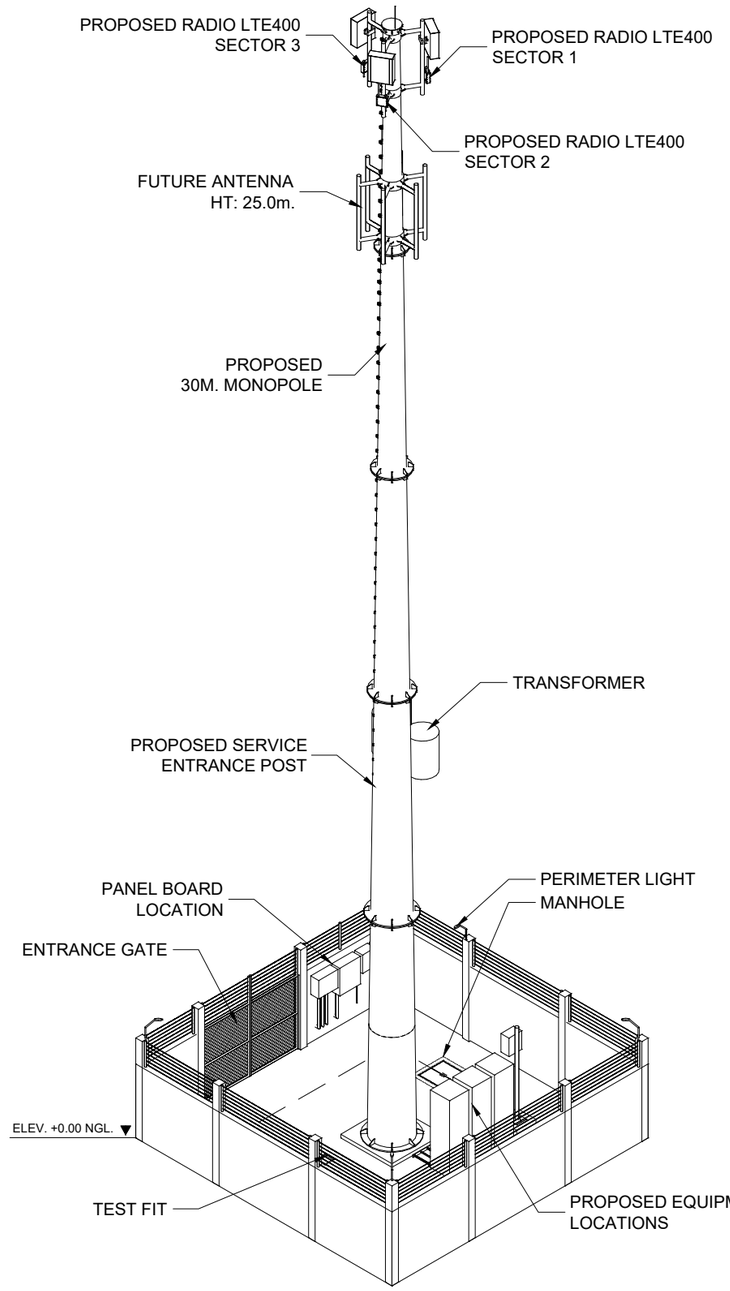
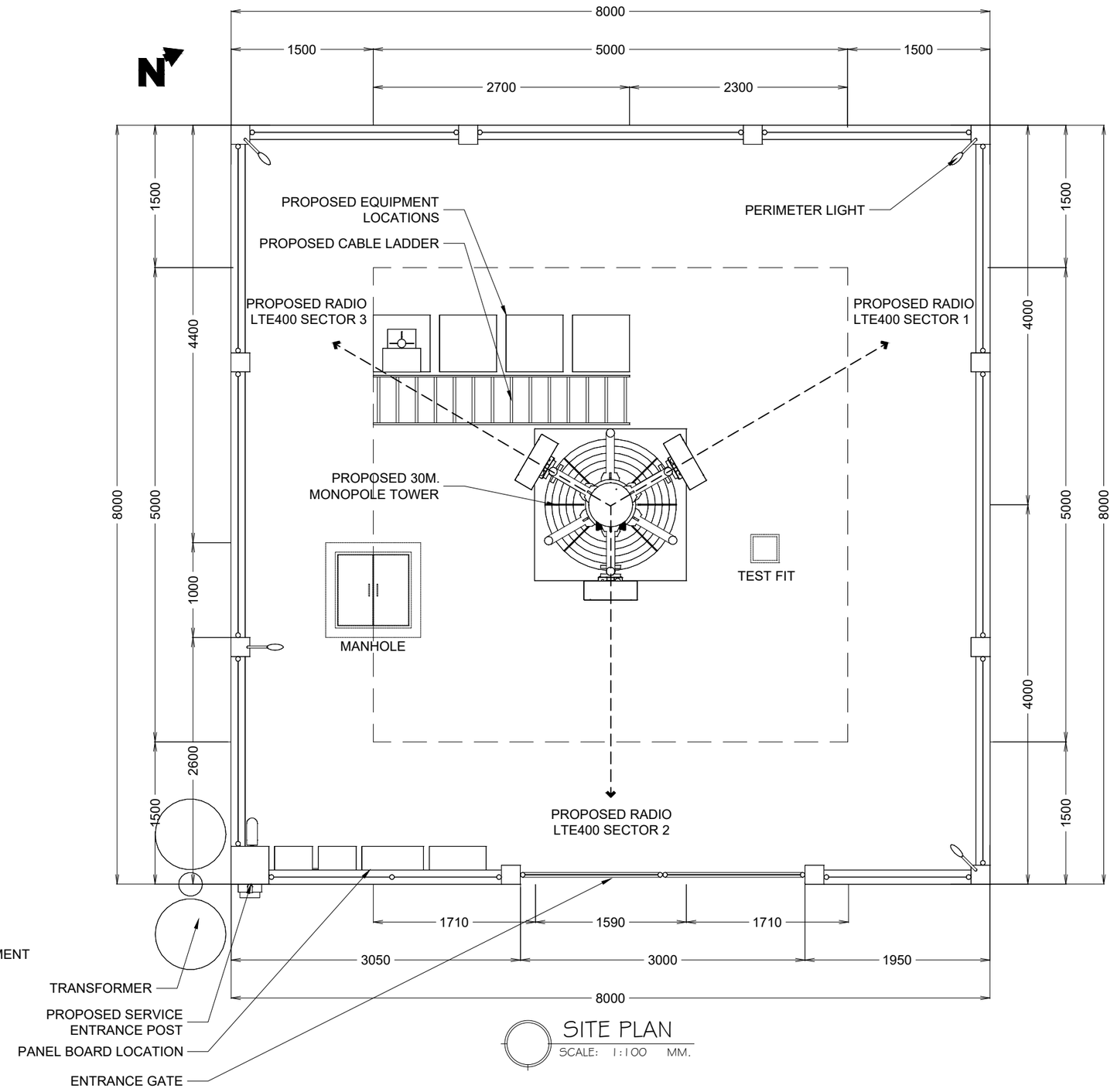


TOWER PLAN

SCALE: NTS.



SITE PLAN

SCALE: 1:100 MM.

OPERATOR: Telecommunications Technology Solutions Inc	OLIVER TUNGOL ENGINEER	SITE PLAN				AUTHORIZED REPRESENTATIVE:	NO.	DATE	DESCRIPTION	SHEET CONTENT:	SHEET NO.:
	PRC NO.: / TIN NO.:										
	DATE ISSUED: / PTR NO.:	ADDRESS:								DRAWING BY.: JOMAR DOGILLO	REV
	PLACE ISSUED:	LONGITUDE:	LATITUDE:	DRAWN	CHECKD	APPRVD					

TELECOMMUNICATIONS TECHNOLOGY SOLUTIONS INCORPORATED (TELTECH)

NATIONWIDE 5G STANDALONE (SA) FWA PROGRAM (BAND n50) — NTC DEMO / TRIAL SYSTEM

SHEET 2: GENERAL STRUCTURAL DESIGN NOTES, PHILIPPINE WIND LOAD MAP & TOWER PARAMETERS

1. PHILIPPINE BASIC WIND SPEED REGIMES (NSCP 2015 / PAGASA HISTORICAL TYPHOON RECURRENCE)

Philippine Wind Zone (NSCP 2015 7th Ed. Sec. 207)	Ultimate Basic Wind Speed (V) (3-sec gust, 50-yr return)	Nominal Wind Speed (Vasd)	Geographic Coverage & Provinces
ZONE 1 (Typhoon Track / Eastern Seaboard)	**315 km/h (87.5 m/s)**	245 km/h (68.0 m/s)	Batanes, Cagayan, Isabela, Aurora, Quezon, Bicol Region, Samar, Leyte, Dinagat, Surigao
ZONE 2 (Central/West Luzon, Visayas)	**270 km/h (75.0 m/s)**	210 km/h (58.3 m/s)	**NCR, Pampanga, Bulacan, Tarlac, Bataan, Nueva Ecija, CALABARZON, Iloilo, Cebu**
ZONE 3 (Inland Mindanao & Palawan)	**200 km/h (55.6 m/s)**	155 km/h (43.1 m/s)	Palawan, Zamboanga Peninsula, Davao, General Santos, Maguindanao, Sulu, Tawi-Tawi

2. TOWER OPERATIONAL & SURVIVABLE WIND LIMIT STATES AND DEFLECTION CRITERIA (TIA-222-H / NSCP 2015)

Design Limit State / Parameter	Standard Threshold / Value	Engineering Criteria & Functional Compliance Rationale
Survivable Wind Speed (Vsurvival)	**270 km/h to 315 km/h** **(3-sec gust, Ultimate Limit State / LRFD)**	Full structural survival under maximum typhoon loading with all antennas, RRUs, and mounts installed. No plastic collapse or foundation overturning.
Operational Wind Speed (Voperational)	**140 km/h to 160 km/h** **(40.0 m/s to 44.4 m/s, Serviceability Limit State)**	Structure ensures continuous 5G active beamforming performance, MIMO rank-4 transmission, and backhaul link availability (≥ 99.995%).
Allowable Twist (θtwist / Torsional Rotation)	≤ **±0.5° (30 arcmin) for MW / Backhaul** ≤ **±1.0° (60 arcmin) for 5G Macro Sector**	Maximum torsional rotation about vertical mast axis under operational wind speed (140-160 km/h) per TIA-222-H Section 2.8.
Allowable Sway / Tilt (θtilt)	≤ **±1.0° (60 arcmin)** **at top antenna mounting centerline (30.0m AGL)**	Angular deflection of vertical mast centerline under operational wind force. Precludes beam pattern overshoot outside target barangay cluster.
Allowable Verticality (Tower Plumbness)	≤ **1 : 500 (0.20% of Tower Height H)** **(Max 60 mm for 30m tower; 80 mm for 40m tower)**	Out-of-plumb tolerance throughout full mast height measured under still air / dead load. Tolerance between consecutive slip-joints ≤ 1:1000 (0.10%).
Top Horizontal Service Deflection	≤ **H / 100 (Max 300 mm at 30.0m AGL)**	Maximum lateral deflection at top of mast under serviceability wind load.

3. CIVIL, STRUCTURAL & ELECTRICAL MATERIAL SPECIFICATIONS

Item / Component	Standard Engineering Specification	Item / Component	Standard Engineering Specification
Monopole Shaft Steel	ASTM A572 Grade 50 (Fy = 345 MPa / 50 ksi, Fu = 450 MPa). 12/18-sided polygon.	**Concrete Strength (fc)**	fc ≥ 28 MPa (4,000 psi) @ 28 days. Class A-1 sulfate-resistant mix.
Slip-Joint Overlap	Minimum 1.5 x inside mating diameter per section joint.	**Reinforcing Steel Rebar**	ASTM A615 / PNS 49 Grade 60 (Fy = 414 MPa / 60 ksi) deformed bars.
Corrosion Protection	Hot-dip galvanized per ASTM A123 / ISO 1461 (≥ 85 µm zinc, ≥ 610 g/m²).	**Anchor Bolts**	ASTM F1554 Grade 55/105 with double heavy-hex leveling and lock nuts.
Compound Footprint	8.0m x 8.0m (64 m²) gravel-surfaced compound with geotextile barrier.	**Lightning & Grounding**	Franklin air terminal (≥ 2.0m above top antenna), ground loop ≤ 5.0 Ohms.

OPERATOR / APPLICANT: Telecommunications Technology Solutions Inc. **NETWORK PARTNER:** Comclark Network and Technology Corp.	**ENGINEER:** JOMAR DOGILLO PRC No.: _____ PTR No.: _____	**AUTHORIZED REPRESENTATIVE:** **OLIVER TUNGOL** Lead Technical Architect & Tender Secretariat	**SHEET CONTENT:** TOWER & SITE STRUCTURAL DESIGN PARAMETERS, PHILIPPINE WIND MAP & DEFLECTION CRITERIA	**SCALE:** NTS **DATE:** October 2026 **SHEET NO.:** **A-2 / 02**